

Ex.no:4b	EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

AIM

To perform data inspection, filtering, sorting, grouping, aggregation, descriptive statistical analysis, and correlation analysis on the **California Housing Dataset** using Pandas, and to visualize the analytical results using Matplotlib and Seaborn.

ALGORITHM

Algorithm 1: Dataset Inspection

1. Import the California Housing Dataset.
2. Display the first few records of the dataset.
3. Check the dimensions of the dataset.
4. Display the data types of the attributes.
5. Generate a statistical summary of the numerical attributes.

Algorithm 2: Filtering Data

1. Select the required attribute such as Median House Value or Median Income.
2. Specify a suitable filtering condition.
3. Apply the condition to the dataset.
4. Display the filtered records.

Algorithm 3: Sorting Data

1. Select the attribute to be sorted such as Median House Value.
2. Apply the `sort_values()` function.
3. Specify the sorting order.
4. Display the sorted dataset.

Algorithm 4: Statistical Analysis

1. Select the required numerical attribute such as Median House Value.
2. Calculate the mean.
3. Calculate the median.
4. Calculate the mode.
5. Calculate the standard deviation.
6. Display the calculated statistical measures.

Algorithm 5: Grouping and Aggregation

1. Select a categorical attribute such as Ocean Proximity.
2. Apply the `groupby()` function.
3. Select the required numerical attribute such as Median House Value.
4. Calculate an aggregate measure such as mean.
5. Display the grouped results.

Algorithm 6: Correlation Analysis

1. Select the required numerical attributes such as Median Income, Total Rooms, Population, and Median House Value.
2. Apply the `corr()` function.
3. Generate the correlation matrix.
4. Analyze the correlation coefficient between the selected attributes.
5. Visualize the relationship using appropriate graphs.

CODE

```
# =====  
# EX NO: 4b  
# EXPLORATORY DATA ANALYSIS  
# DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS  
# AND CORRELATION  
# =====  
  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns  
import os  
  
# =====  
# LOAD DATASET  
# =====  
  
path = "/content"  
  
print("Dataset Path:")  
print(path)  
  
csv_files = [f for f in os.listdir(path) if f.endswith(".csv")]  
  
print("\nCSV Files Found:")  
print(csv_files)  
  
csv_file = "housing prices.csv"  
  
print("\nSelected CSV File:")  
print(csv_file)
```

```

df = pd.read_csv(
    os.path.join(path, csv_file),
    low_memory=False
)

print("\nDataset Loaded Successfully")

# =====
# PROGRAM 1 - DATA INSPECTION
# =====

print("\nFirst 5 Records")
display(df.head())

print("\nLast 5 Records")
display(df.tail())

# =====
# PROGRAM 2 - SHAPE
# =====

print("\nShape of Dataset")
print(df.shape)

print("\nRows =", df.shape[0])
print("Columns =", df.shape[1])

# =====
# PROGRAM 3 - DATA TYPES
# =====

print("\nData Types")
print(df.dtypes)

print("\nDataset Information")
df.info()

# =====
# PROGRAM 4 - FILTERING
# Houses with Median House Value Greater Than 300000
# =====

high_value = df[df["median_house_value"] > 300000]

print("\nRecords with Median House Value Greater Than 300000")

```

```

display(
  high_value[
    [
      "longitude",
      "latitude",
      "median_income",
      "total_rooms",
      "median_house_value",
      "ocean_proximity"
    ]
  ].head(10)
)

# =====
# PROGRAM 5 - FILTER HIGH MEDIAN INCOME
# =====

high_income = df[df["median_income"] > 5]

print("\nRecords with Median Income Greater Than 5")

display(
  high_income[
    [
      "longitude",
      "latitude",
      "median_income",
      "total_rooms",
      "median_house_value",
      "ocean_proximity"
    ]
  ].head(10)
)

# =====
# PROGRAM 6 - SORTING
# Sort by Median House Value
# =====

sorted_df = df.sort_values(
  by="median_house_value",
  ascending=False
)

print("\nTop Records by Median House Value")

display(
  sorted_df[
    [

```

```

        "longitude",
        "latitude",
        "median_income",
        "total_rooms",
        "median_house_value",
        "ocean_proximity"
    ]
].head(10)
)

```

```

# =====
# PROGRAM 7 - BAR CHART
# TOP 10 RECORDS BY HOUSE VALUE
# =====

```

```

top10 = sorted_df.head(10)

plt.figure(figsize=(10, 5))

plt.bar(
    range(len(top10)),
    top10["median_house_value"]
)

plt.xticks(
    range(len(top10)),
    top10["ocean_proximity"],
    rotation=45
)

plt.xlabel("Ocean Proximity")
plt.ylabel("Median House Value")

plt.title("Top 10 Records by Median House Value")

plt.show()

```

```

# =====
# PROGRAM 8 - STATISTICAL ANALYSIS
# =====

```

```

print("\nMean Median House Value =")
print(df["median_house_value"].mean())

print("\nMedian Median House Value =")
print(df["median_house_value"].median())

print("\nMode Median House Value =")
print(df["median_house_value"].mode())

```

```
print("\nStandard Deviation =")
print(df["median_house_value"].std())
```

```
print("\nMinimum Median House Value =")
print(df["median_house_value"].min())
```

```
print("\nMaximum Median House Value =")
print(df["median_house_value"].max())
```

```
# =====
# PROGRAM 9 - DESCRIPTIVE STATISTICS
# =====
```

```
print("\nDescriptive Statistics")
```

```
print(
    df[
        "total_rooms",
        "total_bedrooms",
        "population",
        "households",
        "median_income",
        "median_house_value"
    ]
    .describe()
)
```

```
# =====
# PROGRAM 10 - GROUPING
# Average House Value by Ocean Proximity
# =====
```

```
group = (
    df.groupby("ocean_proximity")["median_house_value"]
    .mean()
    .sort_values(ascending=False)
)
```

```
print("\nAverage House Value by Ocean Proximity")
```

```
print(group)
```

```
# =====
# VISUALIZATION
# =====
```

```
group.plot(
    kind="bar",
    figsize=(8, 5)
)

plt.xlabel("Ocean Proximity")
plt.ylabel("Average House Value")

plt.title("Average House Value by Ocean Proximity")

plt.xticks(rotation=45)

plt.show()
```

```
# =====
# PROGRAM 11 - AGGREGATION
# =====
```

```
summary = df.groupby("ocean_proximity").agg({

    "median_house_value": "mean",

    "median_income": "mean",

    "total_rooms": "mean",

    "total_bedrooms": "mean",

    "population": "mean",

    "households": "mean"

})
```

```
print("\nOcean Proximity-wise Aggregation")
```

```
print(summary)
```

```
# =====
# PROGRAM 12 - CORRELATION
# =====
```

```
numeric = df.select_dtypes(
    include=np.number
)
```

```
corr = numeric.corr()
```

```

print("\nCorrelation Matrix")

print(corr)

# =====
# PROGRAM 13 - CORRELATION HEATMAP
# =====

plt.figure(figsize=(12, 8))

sns.heatmap(
    corr,
    annot=False,
    cmap="coolwarm"
)

plt.title(
    "California Housing Dataset Correlation Heatmap"
)

plt.show()

# =====
# PROGRAM 14 - SCATTER PLOT
# Median Income vs Median House Value
# =====

plt.figure(figsize=(7, 5))

plt.scatter(
    df["median_income"],
    df["median_house_value"]
)

plt.xlabel("Median Income")

plt.ylabel("Median House Value")

plt.title(
    "Median Income vs Median House Value"
)

plt.show()

# =====
# PROGRAM 15 - HISTOGRAM
# Median House Value Distribution
# =====

```



```
plt.figure(figsize=(7, 5))

plt.hist(
    df["median_house_value"].dropna(),
    bins=30
)

plt.xlabel("Median House Value")

plt.ylabel("Frequency")

plt.title(
    "Distribution of Median House Values"
)

plt.show()
```

```
# =====
# PROGRAM 16 - BOXPLOT
# =====
```

```
plt.figure(figsize=(7, 5))

sns.boxplot(
    x=df["median_house_value"].dropna()
)

plt.title(
    "Median House Value Boxplot"
)

plt.xlabel("Median House Value")

plt.show()
```

```
# =====
# PROGRAM 17 - VALUE COUNTS
# =====
```

```
print("\nOcean Proximity Counts")

print(
    df["ocean_proximity"].value_counts()
)
```

```
# =====
# PROGRAM 18 - UNIQUE VALUES
```

```

# =====

print("\nUnique Ocean Proximity Values")

print(
    df["ocean_proximity"].unique()
)

# =====
# PROGRAM 19 - TOP 10 RECORDS BY MEDIAN INCOME
# =====

print("\nTop 10 Records by Median Income")

top_income = (
    df.sort_values(
        "median_income",
        ascending=False
    )
    .head(10)
)

display(
    top_income[
        [
            "longitude",
            "latitude",
            "median_income",
            "median_house_value",
            "ocean_proximity"
        ]
    ]
)

# =====
# PROGRAM 20 - TOP 10 RECORDS BY TOTAL ROOMS
# =====

print("\nTop 10 Records by Total Rooms")

top_rooms = (
    df.sort_values(
        "total_rooms",
        ascending=False
    )
    .head(10)
)

display(

```

```

top_rooms[
    [
        "longitude",
        "latitude",
        "total_rooms",
        "total_bedrooms",
        "median_house_value",
        "ocean_proximity"
    ]
]
)

# =====
# PROGRAM 21 - HOUSE VALUE BY OCEAN PROXIMITY
# =====

print("\nHouse Value by Ocean Proximity")

ocean_value = (
    df.groupby("ocean_proximity")
    ["median_house_value"]
    .sum()
    .sort_values(ascending=False)
)

print(ocean_value)

# =====
# PROGRAM 22 - AVERAGE INCOME BY OCEAN PROXIMITY
# =====

print("\nAverage Median Income by Ocean Proximity")

income_ocean = (
    df.groupby("ocean_proximity")
    ["median_income"]
    .mean()
    .sort_values(ascending=False)
)

print(income_ocean)

# =====
# PROGRAM 23 - AVERAGE ROOMS BY OCEAN PROXIMITY
# =====

print("\nAverage Total Rooms by Ocean Proximity")

```

```
rooms_ocean = (  
    df.groupby("ocean_proximity")  
    ["total_rooms"]  
    .mean()  
    .sort_values(ascending=False)  
)
```

```
print(rooms_ocean)
```

```
# =====  
# PROGRAM 24 - GEOGRAPHICAL SCATTER PLOT  
# =====
```

```
plt.figure(figsize=(8, 6))
```

```
plt.scatter(  
    df["longitude"],  
    df["latitude"],  
    s=5,  
    alpha=0.5  
)
```

```
plt.xlabel("Longitude")
```

```
plt.ylabel("Latitude")
```

```
plt.title(  
    "Geographical Distribution of Housing Records"  
)
```

```
plt.show()
```

```
# =====  
# PROGRAM 25 - INCOME AND HOUSE VALUE SCATTER PLOT  
# =====
```

```
plt.figure(figsize=(8, 6))
```

```
plt.scatter(  
    df["median_income"],  
    df["median_house_value"],  
    alpha=0.4  
)
```

```
plt.xlabel("Median Income")
```

```
plt.ylabel("Median House Value")
```

```
plt.title(  
    "Income and House Value Distribution"
```

```

    "Relationship Between Median Income and House Value"
)

plt.show()

# =====
# PROGRAM 26 - POPULATION VS HOUSE VALUE
# =====

plt.figure(figsize=(8, 6))

plt.scatter(
    df["population"],
    df["median_house_value"],
    alpha=0.4
)

plt.xlabel("Population")

plt.ylabel("Median House Value")

plt.title(
    "Population vs Median House Value"
)

plt.show()

# =====
# PROGRAM 27 - TOP OCEAN PROXIMITY CATEGORY
# =====

print("\nMost Common Ocean Proximity Category")

print(
    df["ocean_proximity"]
    .value_counts()
    .head(1)
)

# =====
# PROGRAM 28 - FINAL STATISTICAL REPORT
# =====

print("\n=====")
print("CALIFORNIA HOUSING EDA REPORT")
print("=====")

print(

```

```
"Dataset Shape :",
df.shape
)

print(
    "Total Records :",
    len(df)
)

print(
    "Number of Ocean Proximity Categories :",
    df["ocean_proximity"].nunique()
)

print(
    "Average Median House Value :",
    df["median_house_value"].mean()
)

print(
    "Minimum Median House Value :",
    df["median_house_value"].min()
)

print(
    "Maximum Median House Value :",
    df["median_house_value"].max()
)

print(
    "Average Median Income :",
    df["median_income"].mean()
)

print(
    "Average Total Rooms :",
    df["total_rooms"].mean()
)

print(
    "Average Population :",
    df["population"].mean()
)

print(
    "Correlation Matrix Generated"
)

print(
    "All EDA Visualizations Generated"
)
```

```
print("=====")

# =====
# END OF EXPERIMENT
# =====

print("\nEXPERIMENT COMPLETED SUCCESSFULLY")
```

OUTPUT:

Dataset Path:

/content

CSV Files Found:

['housing prices.csv']

Selected CSV File:

housing prices.csv

Dataset Loaded Successfully

First 5 Records

	long itude	latitude	housing_m edian_age	total_ rooms	total_be drooms	popu lation	house holds	median _incom e	median_h ouse_value	ocean_p roximity
0	-122.23	37.88	41.0	880.0	129.0	322.0	126.0	8.3252	452600.0	NEAR BAY
1	-122.22	37.86	21.0	7099.0	1106.0	2401.0	1138.0	8.3014	358500.0	NEAR BAY
2	-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100.0	NEAR BAY

	long itude	lati tude	housing_m edian_age	total_ rooms	total_be drooms	popu lation	house holds	median _incom e	median_h ouse_value	ocean_p roximity
3	- 122. 25	37. 85	52.0	1274. 0	235.0	558.0	219.0	5.6431	341300.0	NEAR BAY
4	- 122. 25	37. 85	52.0	1627. 0	280.0	565.0	259.0	3.8462	342200.0	NEAR BAY

Last 5 Records

	long itude	lati tude	housing_ median_a ge	total_ room s	total_b edroom s	popu lation	hous ehold s	median _incom e	median_h ouse_valu e	ocean_p roximit y
20 63 5	- 121. 09	39. 48	25.0	1665. 0	374.0	845.0	330.0	1.5603	78100.0	INLAN D
20 63 6	- 121. 21	39. 49	18.0	697.0	150.0	356.0	114.0	2.5568	77100.0	INLAN D
20 63 7	- 121. 22	39. 43	17.0	2254. 0	485.0	1007. 0	433.0	1.7000	92300.0	INLAN D
20 63 8	- 121. 32	39. 43	18.0	1860. 0	409.0	741.0	349.0	1.8672	84700.0	INLAN D
20 63 9	- 121. 24	39. 37	16.0	2785. 0	616.0	1387. 0	530.0	2.3886	89400.0	INLAN D

Shape of Dataset

(20640, 10)

Rows = 20640

Columns = 10

Data Types


```
longitude      float64
latitude        float64
housing_median_age  float64
total_rooms      float64
total_bedrooms    float64
population       float64
households       float64
median_income     float64
median_house_value float64
ocean_proximity   object
dtype: object
```

Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 20640 entries, 0 to 20639
```

```
Data columns (total 10 columns):
```

#	Column	Non-Null Count	Dtype
0	longitude	20640 non-null	float64
1	latitude	20640 non-null	float64
2	housing_median_age	20640 non-null	float64
3	total_rooms	20640 non-null	float64
4	total_bedrooms	20433 non-null	float64
5	population	20640 non-null	float64
6	households	20640 non-null	float64
7	median_income	20640 non-null	float64
8	median_house_value	20640 non-null	float64

9 ocean_proximity 20640 non-null object

dtypes: float64(9), object(1)

memory usage: 1.6+ MB

Records with Median House Value Greater Than 300000

	longitude	latitude	median_income	total_rooms	median_house_value	ocean_proximity
0	-122.23	37.88	8.3252	880.0	452600.0	NEAR BAY
1	-122.22	37.86	8.3014	7099.0	358500.0	NEAR BAY
2	-122.24	37.85	7.2574	1467.0	352100.0	NEAR BAY
3	-122.25	37.85	5.6431	1274.0	341300.0	NEAR BAY
4	-122.25	37.85	3.8462	1627.0	342200.0	NEAR BAY
89	-122.27	37.80	1.2434	249.0	500001.0	NEAR BAY
103	-122.25	37.81	2.5795	2155.0	350000.0	NEAR BAY
104	-122.26	37.81	2.8406	5871.0	335700.0	NEAR BAY
105	-122.24	37.82	4.9306	1509.0	313400.0	NEAR BAY
118	-122.23	37.84	5.8596	2515.0	327600.0	NEAR BAY

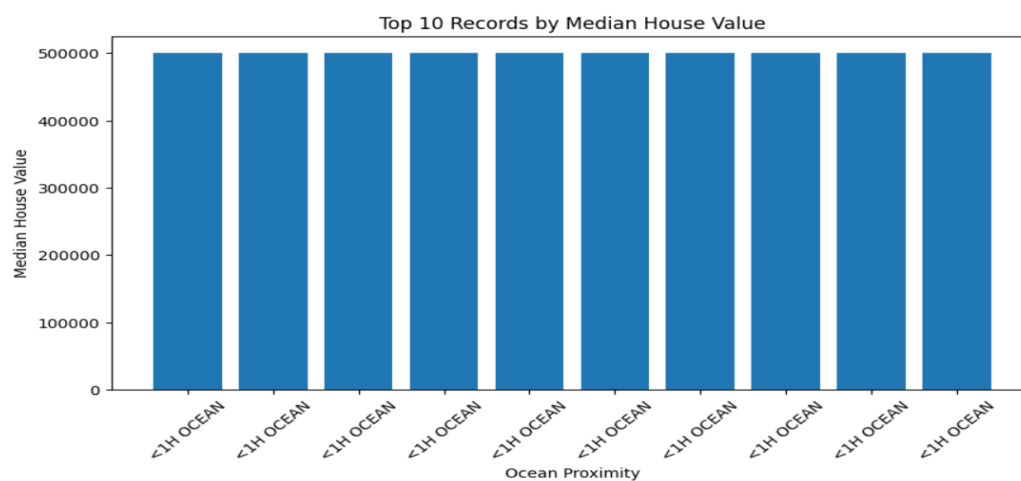
Records with Median Income Greater Than 5

	longitude	latitude	median_income	total_rooms	median_house_value	ocean_proximity
0	-122.23	37.88	8.3252	880.0	452600.0	NEAR BAY
1	-122.22	37.86	8.3014	7099.0	358500.0	NEAR BAY
2	-122.24	37.85	7.2574	1467.0	352100.0	NEAR BAY
3	-122.25	37.85	5.6431	1274.0	341300.0	NEAR BAY
61	-122.29	37.82	6.1183	135.0	75000.0	NEAR BAY
110	-122.25	37.82	5.0917	1424.0	262500.0	NEAR BAY
118	-122.23	37.84	5.8596	2515.0	327600.0	NEAR BAY
119	-122.23	37.84	5.2868	3175.0	347600.0	NEAR BAY
120	-122.24	37.83	5.9560	2576.0	366100.0	NEAR BAY

	longitude	latitude	median_income	total_rooms	median_house_value	ocean_proximity
122	-122.23	37.85	6.3434	2800.0	373600.0	NEAR BAY

Top Records by Median House Value

	longitude	latitude	median_income	total_rooms	median_house_value	ocean_proximity
5253	-118.49	34.11	13.2935	6603.0	500001.0	<1H OCEAN
5254	-118.48	34.07	10.7937	4767.0	500001.0	<1H OCEAN
5255	-118.48	34.07	8.5153	3351.0	500001.0	<1H OCEAN
5256	-118.48	34.07	12.8665	4042.0	500001.0	<1H OCEAN
5257	-118.49	34.06	15.0001	2861.0	500001.0	<1H OCEAN
5258	-118.49	34.07	13.5728	2929.0	500001.0	<1H OCEAN
5259	-118.51	34.11	13.9470	9013.0	500001.0	<1H OCEAN
5260	-118.50	34.05	15.0000	1487.0	500001.0	<1H OCEAN
5261	-118.53	34.09	8.1888	5477.0	500001.0	<1H OCEAN
5262	-118.52	34.05	4.8250	1814.0	500001.0	<1H OCEAN



Mean Median House Value =

206855.81690891474

Median Median House Value =

179700.0

Mode Median House Value =

0 500001.0

Name: median_house_value, dtype: float64

Standard Deviation =

115395.6158744132

Minimum Median House Value =

14999.0

Maximum Median House Value =

500001.0

Descriptive Statistics

	total_rooms	total_bedrooms	population	households \
count	20640.000000	20433.000000	20640.000000	20640.000000
mean	2635.763081	537.870553	1425.476744	499.539680
std	2181.615252	421.385070	1132.462122	382.329753
min	2.000000	1.000000	3.000000	1.000000
25%	1447.750000	296.000000	787.000000	280.000000

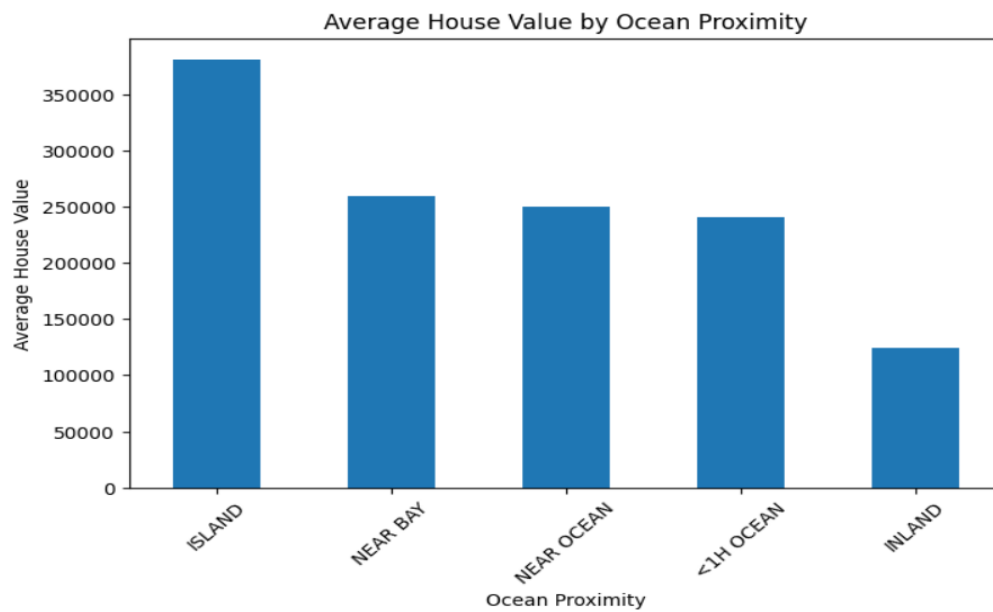
50%	2127.000000	435.000000	1166.000000	409.000000
75%	3148.000000	647.000000	1725.000000	605.000000
max	39320.000000	6445.000000	35682.000000	6082.000000

	median_income	median_house_value
count	20640.000000	20640.000000
mean	3.870671	206855.816909
std	1.899822	115395.615874
min	0.499900	14999.000000
25%	2.563400	119600.000000
50%	3.534800	179700.000000
75%	4.743250	264725.000000
max	15.000100	500001.000000

Average House Value by Ocean Proximity

ocean_proximity

ISLAND	380440.000000
NEAR BAY	259212.311790
NEAR OCEAN	249433.977427
<1H OCEAN	240084.285464
INLAND	124805.392001



Ocean Proximity-wise Aggregation

```

median_house_value median_income total_rooms \
ocean_proximity

<1H OCEAN      240084.285464    4.230682 2628.343586
INLAND         124805.392001    3.208996 2717.742787
ISLAND         380440.000000    2.744420 1574.600000
NEAR BAY       259212.311790    4.172885 2493.589520
NEAR OCEAN     249433.977427    4.005785 2583.700903

```

```

total_bedrooms population households
ocean_proximity

<1H OCEAN      546.539185 1520.290499 517.744965
INLAND         533.881619 1391.046252 477.447565
ISLAND         420.400000  668.000000 276.600000
NEAR BAY       514.182819 1230.317467 488.616157
NEAR OCEAN     538.615677 1354.008653 501.244545

```

Correlation Matrix

	longitude	latitude	housing_median_age	total_rooms	\
longitude	1.000000	-0.924664	-0.108197	0.044568	
latitude	-0.924664	1.000000	0.011173	-0.036100	
housing_median_age	-0.108197	0.011173	1.000000	-0.361262	
total_rooms	0.044568	-0.036100	-0.361262	1.000000	
total_bedrooms	0.069608	-0.066983	-0.320451	0.930380	
population	0.099773	-0.108785	-0.296244	0.857126	
households	0.055310	-0.071035	-0.302916	0.918484	
median_income	-0.015176	-0.079809	-0.119034	0.198050	
median_house_value	-0.045967	-0.144160	0.105623	0.134153	

	total_bedrooms	population	households	median_income	\
longitude	0.069608	0.099773	0.055310	-0.015176	
latitude	-0.066983	-0.108785	-0.071035	-0.079809	
housing_median_age	-0.320451	-0.296244	-0.302916	-0.119034	
total_rooms	0.930380	0.857126	0.918484	0.198050	
total_bedrooms	1.000000	0.877747	0.979728	-0.007723	
population	0.877747	1.000000	0.907222	0.004834	
households	0.979728	0.907222	1.000000	0.013033	
median_income	-0.007723	0.004834	0.013033	1.000000	
median_house_value	0.049686	-0.024650	0.065843	0.688075	

	median_house_value
longitude	-0.045967
latitude	-0.144160
housing_median_age	0.105623
total_rooms	0.134153

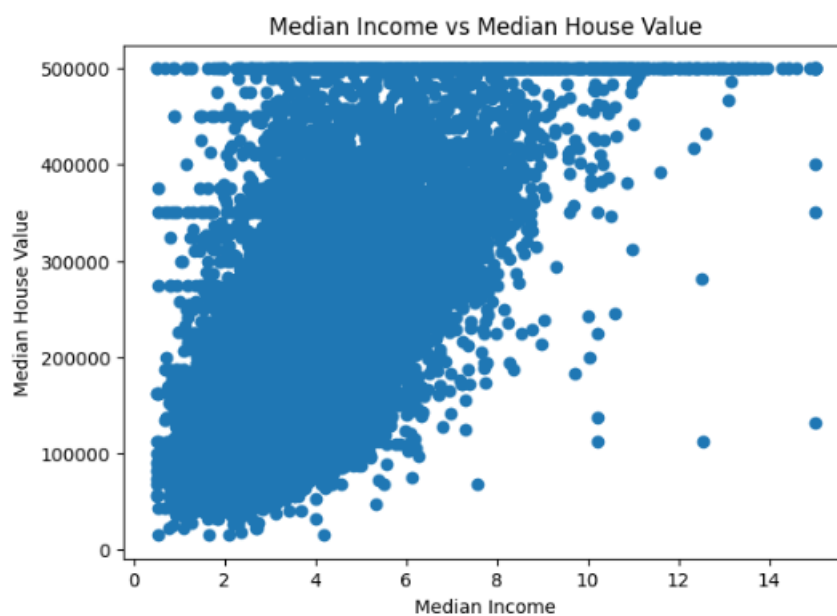
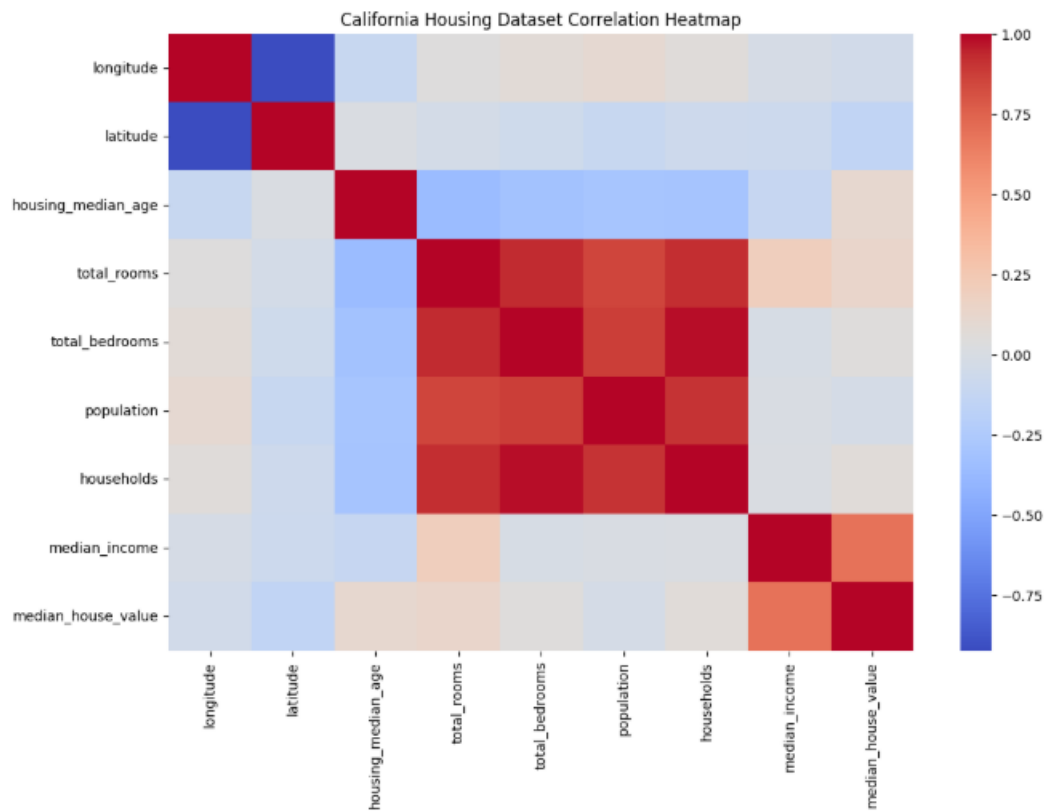
total_bedrooms 0.049686

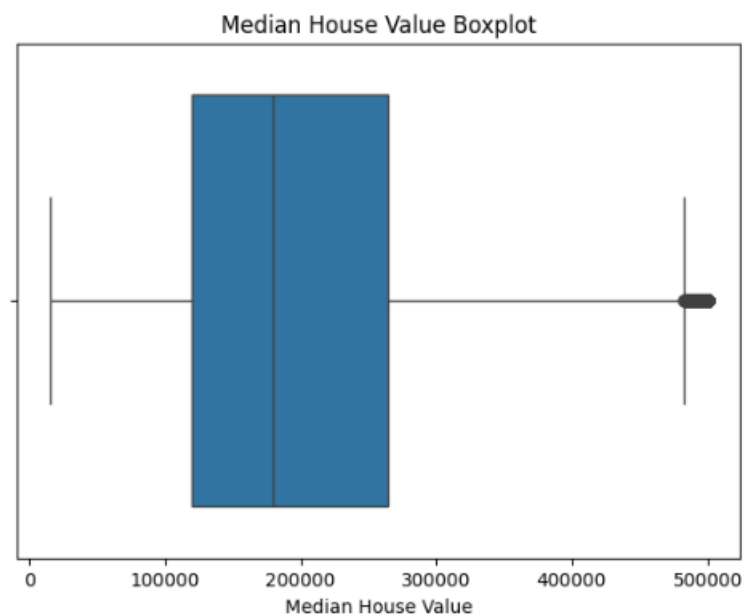
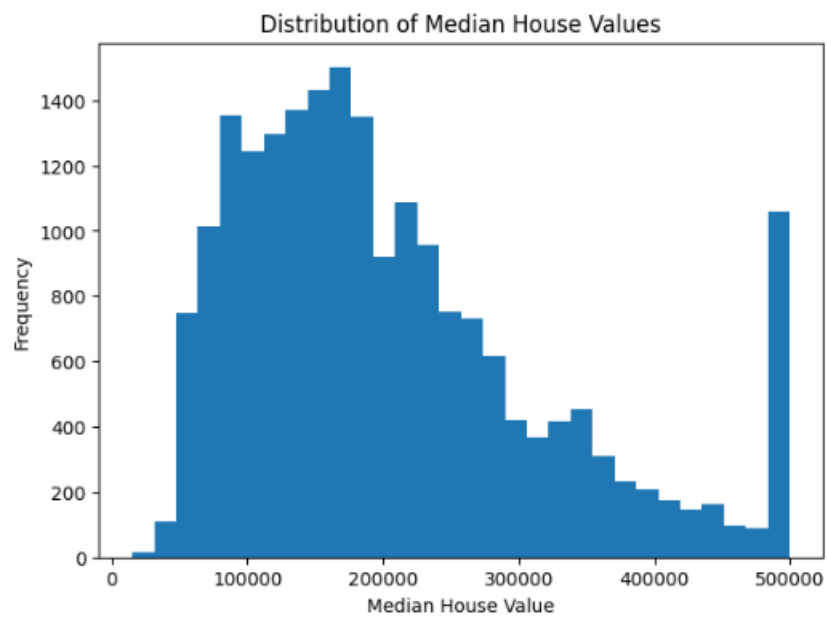
population -0.024650

households 0.065843

median_income 0.688075

median_house_value 1.000000





Name: count, dtype: int64

Unique Ocean Proximity Values

['NEAR BAY' '<1H OCEAN' 'INLAND' 'NEAR OCEAN' 'ISLAND']

Top 10 Records by Median Income

	longitude	latitude	median_income	median_house_value	ocean_proximity
8851	-118.42	34.09	15.0001	500001.0	<1H OCEAN
8850	-118.41	34.09	15.0001	500001.0	<1H OCEAN
8849	-118.40	34.08	15.0001	500001.0	<1H OCEAN
8848	-118.39	34.08	15.0001	500001.0	<1H OCEAN
8852	-118.42	34.08	15.0001	500001.0	<1H OCEAN
15241	-117.23	32.99	15.0001	500001.0	NEAR OCEAN
8878	-118.50	34.04	15.0001	500001.0	<1H OCEAN
8805	-118.34	33.76	15.0001	500001.0	NEAR OCEAN
8847	-118.40	34.09	15.0001	500001.0	<1H OCEAN
16910	-122.36	37.56	15.0001	500001.0	NEAR OCEAN

Top 10 Records by Total Rooms

	longitude	latitude	total_rooms	total_bedrooms	median_house_value	ocean_proximity
13139	-121.44	38.43	39320.0	6210.0	153700.0	INLAND
10309	-117.74	33.89	37937.0	5471.0	366300.0	<1H OCEAN
9880	-121.79	36.64	32627.0	6445.0	118800.0	<1H OCEAN
6057	-117.78	34.03	32054.0	5290.0	253900.0	<1H OCEAN
12201	-117.20	33.58	30450.0	5033.0	174300.0	<1H OCEAN
9019	-118.78	34.16	30405.0	4093.0	399200.0	NEAR OCEAN
12215	-117.12	33.52	30401.0	4957.0	212300.0	<1H OCEAN
922	-121.92	37.53	28258.0	3864.0	451100.0	<1H OCEAN
12623	-121.53	38.48	27870.0	5027.0	212200.0	INLAND
6066	-117.87	34.04	27700.0	4179.0	339700.0	<1H OCEAN

House Value by Ocean Proximity

ocean_proximity

<1H OCEAN 2.193410e+09

INLAND 8.176001e+08

NEAR OCEAN 6.629955e+08

NEAR BAY 5.935962e+08

ISLAND 1.902200e+06

Name: median_house_value, dtype: float64

Average Median Income by Ocean Proximity

ocean_proximity

<1H OCEAN 4.230682

NEAR BAY 4.172885

NEAR OCEAN 4.005785

INLAND 3.208996

ISLAND 2.744420

Name: median_income, dtype: float64

Average Total Rooms by Ocean Proximity

ocean_proximity

INLAND 2717.742787

<1H OCEAN 2628.343586

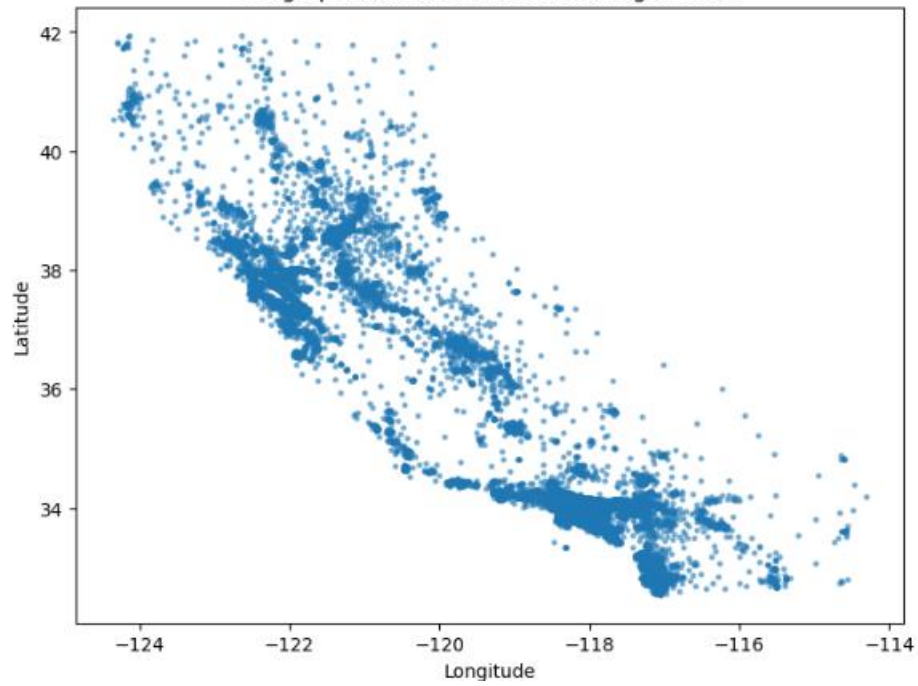
NEAR OCEAN 2583.700903

NEAR BAY 2493.589520

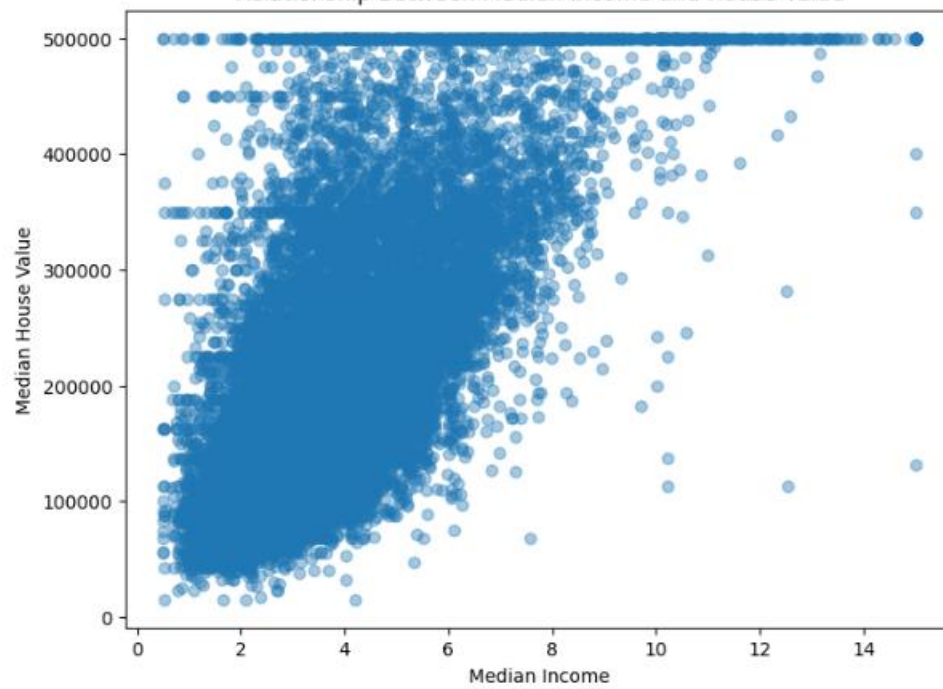
ISLAND 1574.600000

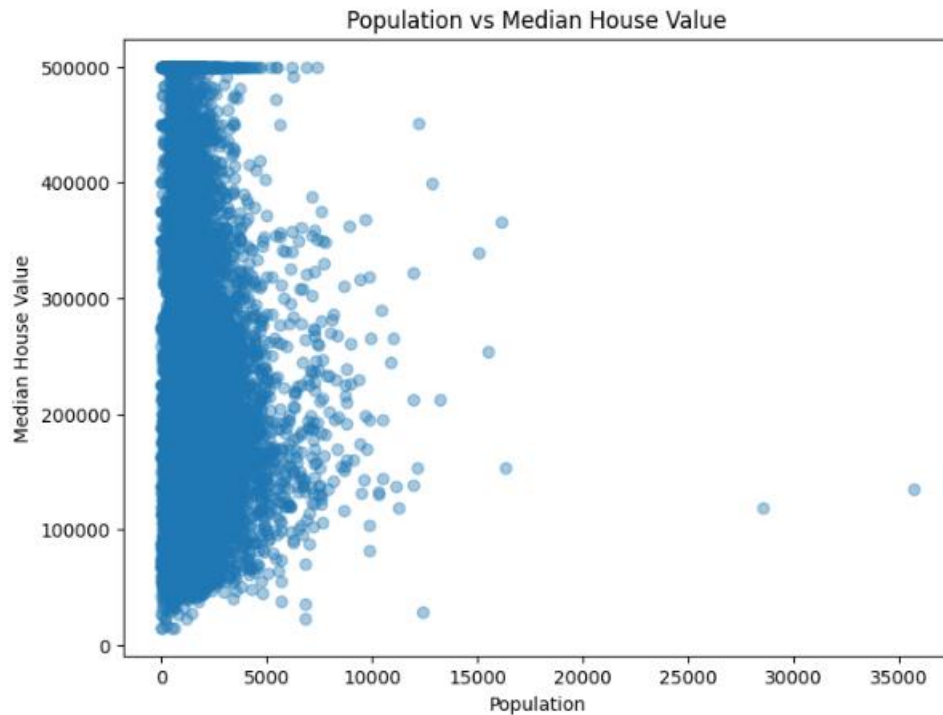
Name: total_rooms, dtype: float64

Geographical Distribution of Housing Records



Relationship Between Median Income and House Value





Most Common Ocean Proximity Category

ocean_proximity

<1H OCEAN 9136

Name: count, dtype: int64

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CALIFORNIA HOUSING EDA REPORT

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Dataset Shape : (20640, 10)

Total Records : 20640

Number of Ocean Proximity Categories : 5

Average Median House Value : 206855.81690891474

Minimum Median House Value : 14999.0

Maximum Median House Value : 500001.0

Average Median Income : 3.8706710029069766

Average Total Rooms : 2635.7630813953488

Average Population : 1425.4767441860465

Correlation Matrix Generated

All EDA Visualizations Generated

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EXPERIMENT COMPLETED SUCCESSFULLY

RESULT